

# Interim Milk-based Tier 1 Phantom SOP

Visions PV .035 · T1-E4\* and T1-E5\* interim stand-ins

## Interim Milk-based Tier 1 Phantom SOP (T1-E4\* / T1-E5\*)

This SOP defines two **interim** Tier 1 phantom experiments — **T1-E4\*** and **T1-E5\*** — built around evaporated milk and a milk-agar gel, intended to keep Tier 1 calibration progressing while the canonical agar-graphite phantoms ([A.2](#), [A.3](#) of [ivus\\_calibration\\_protocol.md](#)) are blocked on reagent procurement (specifically: micrometer-scale flake graphite and a verified-CAS n-propanol or glycerol). Both experiments target the same parameters as the canonical [E4](#) and [E5](#), with looser  $\alpha$  and  $c$  tolerances that must be **flagged in every output's substrate-parameter-sheet entry**.

The asterisk in [T1-E4\\*](#) / [T1-E5\\*](#) indicates an interim derivative of the canonical experiment. Outputs from these experiments are **batch-1 interim priors only** and must be re-derived once the canonical agar-graphite phantoms are available.

| Interim experiment | Replaces (canonical)                                                            | Phantom                                                                                                                                                                                          | What we keep                                                                                                                                                                                                                                                                                                         | What we lose                                                               |
|--------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| T1-E4*             | E4 — Uniform attenuation phantom + partial E7 gain-linearity + partial E5 noise | Beaker of evaporated milk <b>captured at two dilutions</b> (undiluted, 1:1 with degassed distilled water) + the existing E2 wire phantom (submerged briefly in each dilution for $c$ refinement) | TGC schedule, $\alpha$ prior (loose) at two scatterer densities, refined $c$ from wire-phantom TOF (two independent estimates), $\text{gain\_db}$ slider linearity (cross-check on E7), noise $\sigma$ from $\text{gain} = 20$ (cross-check on E5), $\alpha$ -vs-milk-fraction linearity (pipeline self-consistency) | Tight $\alpha$ calibration to $\pm 0.10$ dB/cm/MHz                         |
| T1-E5*             | E5 — Cyst phantom                                                               | Agar + evaporated-milk + glycerin gel + PTFE/dowel-rod-formed cyst voids                                                                                                                         | noise $\sigma$ , $\text{scattering\_resolution\_mm}$ , $\mu 0/\mu 1/\sigma$ priors, $\text{reject\_db}$ , cyst contrast                                                                                                                                                                                              | Recipe-matched $\alpha$ ; speckle-statistic batch-to-batch reproducibility |

The canonical [E7](#) step phantom has **no interim equivalent** in this SOP — it depends on a per-chamber graphite  $\alpha$  ladder that talc and milk-based gels cannot replicate. Defer E7 until the graphite arrives.

All Tier 1 / Tier 1\* experiments are run with the same Volcano s5i console, the same catheter, and at  $22 \pm 1$  °C unless otherwise noted. Speed of sound in degassed water at 22 °C is  **$c_{\text{water}} = 1488$  m/s** (Marczak 1997).

## Common equipment (deltas from the canonical protocol's "Common equipment" table)

| Item                                                | Spec                                                                                                                                                                                               | Tolerance / Notes                                                                                                                  |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Volcano s5i console + Visions PV .035 catheter      | service-mode access                                                                                                                                                                                | Same unit and same catheter S/N as the canonical <a href="#">P_035_PointScatter/</a> dataset                                       |
| Catheter fixture                                    | rigid clamp, vertical orientation preferred so catheter long axis is normal to gravity <i>only</i> if the phantom container is also held vertical; otherwise horizontal — see per-experiment setup | axial alignment $\leq \pm 0.1$ mm; does not occlude the imaging plane                                                              |
| Calibrated thermometer                              | $\pm 0.2$ °C                                                                                                                                                                                       | Log T at the start <b>and end</b> of every session (milk c is $\sim -3$ m/s per °C; temperature stability matters more than usual) |
| Degassed deionized water                            | dissolved $O_2 \leq 4$ ppm                                                                                                                                                                         | Used for the agar dissolution step in T1-E5* only                                                                                  |
| RF capture (preferred) / frame grabber (acceptable) | as per canonical <a href="#">ivus_calibration_protocol.md</a> Common equipment table                                                                                                               | Same as canonical                                                                                                                  |

## Common console settings (reference operating point — match the calibration sheet)

The console settings used in this SOP are the **calibration-sheet reference operating point** from `../p035_visions/volcano_s5i.yaml`, **not** the older E4/E5 settings of "imaging diameter = 16 mm". The sheet was calibrated against `P_035_PointScatter/` at 60 mm imaging diameter and gain slider 54; we capture at that operating point so the extracted parameters can be consumed by the simulator without re-validating Tier 1 at a different diameter.

| Setting                       | T1-E4*                                                                                                                                                                                                                            | T1-E5*                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Imaging mode                  | IVUS B-mode                                                                                                                                                                                                                       | IVUS B-mode                                |
| Imaging diameter              | <b>60 mm</b>                                                                                                                                                                                                                      | <b>60 mm</b>                               |
| Gain slider                   | <b>{20, 35, 50, 68}</b> (50 = canonical E4 reference operating point; 20 = noise-floor anchor; 35 and 68 added to give a 4-point fit on the gain_db(slider) transfer — three points only check linearity, four points detect bow) | <b>{20, 50, 68}</b> (matches canonical E5) |
| TGC sliders                   | <b>All centered</b> (flat / disabled)                                                                                                                                                                                             | All centered                               |
| Acoustic Reference (AR)       | <b>ON</b>                                                                                                                                                                                                                         | <b>ON</b>                                  |
| Speckle reduction / smoothing | OFF                                                                                                                                                                                                                               | OFF                                        |
| Compounding                   | OFF                                                                                                                                                                                                                               | OFF                                        |

If any setting deviates, **stop** — the capture won't be comparable to the calibration sheet.

## Substrate identification

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Every batch must carry a `substrate_id`. T1-E4\* generates **two** substrate IDs per session (one per dilution):

- `tmm-milk-INTERIM-<YYYYMMDD>-undiluted` — Phase A1 + B1 captures.
- `tmm-milk-INTERIM-<YYYYMMDD>-dil1to1` — Phase A2 + B2 captures (1:1 by volume evaporated milk : freshly degassed distilled water).

T1-E5\* generates one ID per gel batch:

- `tmm-milk-agar-INTERIM-<YYYYMMDD>` — the milk-agar-glycerin cyst gel.

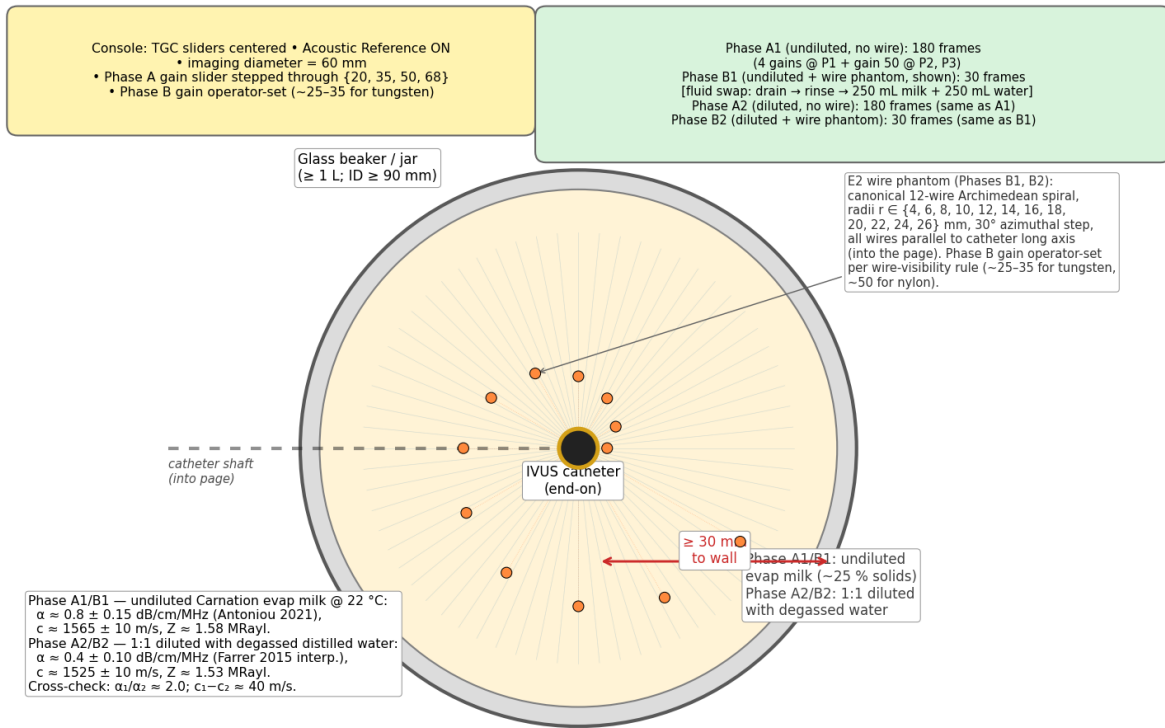
The **INTERIM** token in the substrate ID is intentional: it ensures the parameter-sheet analysis pipeline can quickly identify these as not-yet-canonical and flag them for re-derivation when the canonical phantom arrives.

Every interim parameter row in the substrate parameter sheet must carry `Source: bench, interim milk-* batch <date>, no substitution measurement` in the `Source` column.

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## T1-E4\* — Uniform-fluid milk tank (interim stand-in for E4)

### T1-E4\* — Uniform-fluid milk tank + wire-phantom c-cal (interim stand-in for E4; cross-section view, Phase B1 shown)



**Produces:** interim `processing.tgc_control_points` (cross-validated across two milk dilutions), interim `materials[].attenuation_db_per_cm_mhz` priors at two scatterer densities, interim refined `probe.speed_of_sound_mm_per_us` (from wire-phantom-in-milk acquisitions in both dilutions), interim `processing.gain_db(slider)` cross-check at the four captured gains (partial E7 output), and an interim `processing.noise.sigma` cross-check from the gain = 20 capture (partial E5 output, useful for self-consistency against T1-E5\*). The **two-dilution structure** is the core self-consistency check of T1-E4\*:  $\alpha$  from the two dilutions must scale linearly with milk fraction (expected ratio  $\approx 2:1$ ) and the recovered TGC curve must agree across the two dilutions to within  $\pm 2$  dB over the 30 mm imaging range.

**Substrate ID pattern:** `tmm-milk-INTERIM-<YYYYMMDD>-undiluted` and `tmm-milk-INTERIM-<YYYYMMDD>-dil1to1` (one row per dilution in the substrate parameter sheet).

**Goal.** Use a uniform attenuating *fluid* with literature-known  $c$  and  $\alpha$  to extract a TGC schedule and a loose  $\alpha$  prior, without waiting for the canonical agar-graphite phantom. We capture at **two dilutions** — undiluted Carnation evaporated milk ( $\alpha \approx 0.8$  dB/cm/MHz, Antoniou 2021) and 1:1 with degassed distilled water ( $\alpha \approx 0.4$  dB/cm/MHz, interpolated; brackets the 0.5 dB/cm/MHz soft-tissue target of Goss-Johnston-Dunn 1978/1980). The two-point capture cross-validates the  $\alpha$ -extraction pipeline ( $\alpha$  ratio should be  $\approx 2.0$  by Antoniou's milk-fraction linearity), self-consistency-checks the recovered TGC curve (same console TGC must be recovered from both substrates), and gives independent  $c$  estimates from the wire phantom at two different bulk- $c$  values (~1565 m/s vs ~1525 m/s), catching any systematic TOF-extraction bias.

Evaporated milk is a published tissue-mimicking material — see the References section at the bottom of this SOP for the foundational papers, in particular Madsen-Frank-Dong (1998) for the WARF milk-based gel (the patent that backs the commercial CIRS / ATS milk phantoms), Farrer et al. (2015) for the gelatin + 50/50 water-milk variant (open access, reports  $c \approx 1540$  m/s and  $\alpha \approx 0.4$ – $0.5$  dB/cm/MHz — anchors our diluted phase), and Antoniou et al. (2021) for the agar + evaporated-milk recipe that most closely matches our T1-E5\* gel (open access, reports  $\alpha$  scaling with milk fraction in the

0.3–1.0 dB/cm/MHz range — anchors our undiluted phase at the 0.8 dB/cm/MHz end and the dilution-linearity cross-check).

## Equipment

| Item                               | Spec                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Tolerance                                                                                                                                                                                                                               |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evaporated milk                    | Carnation <b>evaporated</b> milk (or any brand meeting the FDA evaporated-milk standard of identity, ~25 % total solids minimum). <i>Not</i> sweetened condensed milk. <b>Plan 2 cans per session</b> — ~1 can (~350 mL) for the undiluted phase A1/B1 fill and ~0.7 can for the diluted phase A2/B2 fill (250 mL milk + 250 mL degassed water). Do not mix milk between cans of different brand / fat content; if topping up, use the same can.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Best-by date logged; cans opened on the day of the session                                                                                                                                                                              |
| Distilled water (degassed)         | ≥ 300 mL of freshly degassed distilled water, prepared by boiling for 5 min in a covered pot and cooling to room temperature still covered. Used for the 1:1 dilution in phase A2/B2 only. Do not substitute tap water (chlorinated, microbubble-prone) or stored "degassed" water (re-absorbs O <sub>2</sub> within hours).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Dissolved O <sub>2</sub> ≤ 4 ppm (proxy: no visible bubbles forming on the container wall during the 30 min settle)                                                                                                                     |
| Measuring jug / graduated cylinder | 500 mL, ± 5 mL graduations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | For measuring 250 mL milk + 250 mL degassed water for the 1:1 dilution                                                                                                                                                                  |
| Container                          | Tall glass beaker, jar, or plastic food-storage container. <b>All four phases need:</b> ID ≥ 90 mm (so the wire-phantom disc, Ø 80 mm, fits with ≥ 5 mm radial clearance per side) and fill height ≥ 80 mm above the floor (so the catheter element ring sits ≥ 30 mm above the floor and ≥ 30 mm below the milk surface in the imaging plane). The 30 mm clearance requirement applies to the imaging plane around the catheter element ring (so the milk acts as semi-infinite within the imaging fan), <b>not</b> around the wire-phantom assembly — the disc faces sit ±25 mm out of the imaging plane along the catheter long axis, well outside the IVUS slice thickness (~1 mm) and therefore invisible to the transducer. <b>A 9 × 9 × 8 cm plastic food-storage container is known-good for all four phases</b> when paired with a hole-in-the-lid catheter mount. A 1 L Pyrex beaker (ID ≈ 100 mm, height ≈ 145 mm) or a wide-mouth 32 oz Ball mason jar (ID ≈ 95 mm, height ≈ 170 mm) is the preferred fit when on hand. 500 / 600 mL beakers (ID ~78–82 mm) are too narrow for the wire phantom and force the single-wire c-cal fallback. <b>One container is sufficient</b> — drain and rinse with degassed water between the undiluted and diluted phases. | Probe-to-wall ≥ 30 mm in the imaging plane (all phases); disc-to-wall ≥ 5 mm (phases B1/B2 only — the disc itself is axially out of the imaging plane so its near-wall position only matters for mechanical clearance during insertion) |
| Catheter mount                     | Holds catheter horizontal with the element ring submerged near the center of the container; must remain in place across the fluid swap between phases B1 → A2 so that the catheter axial position (P1) is reproducible across dilutions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ≤ ±0.1 mm axial position; ≤ ±0.5 mm radial position before and after fluid swap                                                                                                                                                         |
| c-cal distance marker (primary)    | The existing <b>E2 wire phantom</b> used for canonical Tier 1 PSF work ( <code>hardware/wire_spiral_disc.stl</code> , 12 wires on a 1-turn Archimedean spiral at canonical radii / azimuths {(4, 0°), (6, 30°), (8, 60°), (10, 90°), (12, 120°), (14, 150°), (16, 180°), (18, 210°), (20, 240°), (22, 270°), (24, 300°), (26, 330°)}, all wires parallel to the catheter long axis). Submerged in milk for the c-refinement acquisitions only; the phantom must fit inside the milk container with ≥ 30 mm clearance to walls in the imaging plane. <b>Wire material is determined by the canonical E2 build</b> — if your build uses 30 µm tungsten (the Mie-resonance-avoidance choice for the PV .035), expect Phase B captures to need lower gain (~25–35) than Phase A.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Design radii ± 0.05 mm (calibrated per the canonical E2 fixture)                                                                                                                                                                        |
| c-cal distance marker (fallback)   | A single 0.05 mm tungsten wire (or any reflective ≥ 25 µm wire) suspended vertically through the milk so it crosses the imaging plane at a measured radial distance. Use only if the E2 wire phantom is unavailable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | r measured by digital calliper ± 0.05 mm                                                                                                                                                                                                |
| Calibrated thermometer             | ± 0.2 °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Log T at the start and end of each phase (4 log points minimum)                                                                                                                                                                         |

## Setup procedure

1. **Degas the dilution water.** Boil  $\geq 300$  mL distilled water for 5 min in a covered pot, then cover and cool to room temperature ( $\sim 30$ – $40$  min). Keep the lid on while cooling so the water doesn't re-absorb atmospheric  $O_2$ . This can run in parallel with the rest of setup and Phase A1 — by the time Phase A1/B1 are done, the water should be at room temp and ready for the 1:1 mix.
2. Open one fresh can of evaporated milk and pour it into the container until the fill height is  $\geq 80$  mm above the floor ( $\sim 350$  mL for the  $9 \times 9$  cm container). Top up with milk from a second can if needed to fully submerge the element ring at P1 with  $\geq 30$  mm clearance below the milk surface.
3. Let the container sit, covered with cling film or a clean lid, at room temperature for  $\geq 30$  min to allow any pour-induced bubbles to escape. Log  $T_{\text{room}}$  and  $T_{\text{milk}}$  at the end of equilibration. Do not shake the can or stir the contents — you want a settled, uniform fluid.
4. Mount the catheter horizontally so the element ring is centered in the container,  $\geq 30$  mm from all walls (sides and bottom). Verify by visual inspection / calliper from the outside. **Do not unmount the catheter between phases** — the mount must remain registered to the same P1, P2, P3 positions through the entire session, including the fluid swap between phases B1 and A2.
5. Confirm the console is at the reference operating point per the table above (gain slider will be stepped during the run; start at 50). Take a screen capture / photo of the slider state.

## Acquisition procedure

The capture runs in **four phases**, in order:

| Phase | Substrate                                                  | Captures                                                                                                                                          | Notes                                                                                                                               |
|-------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| A1    | Undiluted evap milk (Carnation, $\sim 25$ % solids)        | 4 gains $\times$ P1 + gain-50 $\times$ P2, P3 = 180 frames                                                                                        | Main TGC + $\alpha$ extraction run; literature $\alpha$ anchor 0.8 dB/cm/MHz (Antoniou 2021)                                        |
| B1    | Undiluted evap milk + E2 wire phantom                      | <b>operator-set gain</b> $\times$ 30 frames (typically 25–35 for tungsten, 50 for nylon — pick on the bench using the wire-visibility rule below) | c refinement at the undiluted bulk-c ( $\sim 1565$ m/s expected)                                                                    |
| A2    | 1:1 evap milk : degassed distilled water (250 mL + 250 mL) | 4 gains $\times$ P1 + gain-50 $\times$ P2, P3 = 180 frames                                                                                        | Cross-validation TGC + $\alpha$ run; literature $\alpha$ anchor 0.4 dB/cm/MHz (Farrer 2015 interp.); same P1/P2/P3 positions as A1  |
| B2    | 1:1 diluted milk + E2 wire phantom                         | operator-set gain $\times$ 30 frames (same gain as Phase B1 if possible — use the operator-set gain from B1, log it in metadata)                  | c refinement at the diluted bulk-c ( $\sim 1525$ m/s expected) — independent check of TOF-extraction pipeline at a different bulk c |

**Phase B gain rule.** Phase B is for c calibration via wire-time-of-flight, not PSF shape — saturated wire peaks are acceptable as long as **peak locations remain unambiguous** (no merging with adjacent peaks or with the catheter ring-down). Set the gain such that:

- the **deepest** clearly-visible wire (target  $r = 26$  mm) is  $\geq 20$  dB above the noise floor, AND
- the **shallowest** visible wire (target  $r = 4$  mm) has  $\leq 95$  % of its peak ring inside the displayed-palette top (i.e. no saturation blob wider than  $\sim 2$  azimuthal bins).

For 30  $\mu$ m tungsten on the PV .035 this typically falls at **gain 25–35** (well below the Phase A reference of 50); for nylon monofilament, gain 50 usually meets both criteria. The operator-set value is logged in the session metadata.

**Why the four-phase structure.** The console TGC and the slider $\rightarrow$ dB transfer are probe properties that must be invariant across substrate; capturing in two substrates with a factor-of-2 difference in  $\alpha$  gives us a redundancy check on both. The recovered TGC curve must match across A1 and A2 to within  $\pm 2$  dB, and the extracted  $\alpha$  values must scale  $\sim 2:1$  (Antoniou 2021 confirms linear  $\alpha$  scaling with milk fraction). Both checks fail loudly if anything in the analysis pipeline has a bias.

Phase B1 and B2 may be skipped if the E2 wire phantom is not on the bench *that day*, but are strongly preferred — they add  $\sim 3$  min each and fold in 5 distance markers at known radii per dilution, plus cross-check TOF extraction at two bulk-c values.

### Phase A1 — multi-gain undiluted-milk capture

1. **Position 1, gain sweep.** With the catheter centered in the beaker and the milk settled, capture **30 frames at each of gains {20, 35, 50, 68}** in ascending order. The catheter must remain stationary between gains — change *only* the gain slider. (Total at P1: 120 frames across 4 gains.)
2. **Position 2, gain 50 only.** Move the catheter  $\geq 5$  mm along its long axis (or move the container vertically by the same distance). Capture **30 frames at gain 50**.
3. **Position 3, gain 50 only.** Repeat. Total Phase A1:  $30 \times 4 + 30 + 30 = 180$  frames (4 gains  $\times$  30 at P1 + 30 each at P2, P3).
4. Return the catheter to P1 (the position must be reproduced for A2).
5. Log T<sub>milk</sub> at the end of Phase A1.

### Phase B1 — wire-phantom c-refinement in undiluted milk

6. Lower the **E2 wire phantom** into the milk so the wires straddle the imaging plane of the catheter. Use the same fixture that holds the wire phantom for the canonical E2 (water) acquisition; do not try to free-hand-suspend it. If your container has a hole-in-the-lid catheter mount, feed the catheter through the disc centre holes ( $\varnothing 2.5$  mm) of the wire phantom on its way into the milk so the assembly hangs from the catheter shaft, with the catheter element ring centred in the 50 mm standoff gap between the two discs. Verify the geometry: - Container ID accommodates the  $\varnothing 80$  mm disc with  $\geq 5$  mm radial clearance per side (e.g., 90 mm ID gives 5 mm; 100 mm gives 10 mm). - **In the imaging plane** (horizontal slice through the catheter element ring): probe-to-wall  $\geq 30$  mm in every direction. With ID  $\geq 90$  mm this is automatic since the catheter is centred and the imaging fan is only 30 mm radius. - **Along the catheter axis** (vertical): the catheter element ring sits  $\geq 30$  mm above the container floor and  $\geq 30$  mm below the milk surface, so the milk extends  $\geq 30$  mm in every direction around the imaging plane. The disc faces at  $\pm 25$  mm out of the imaging plane are *outside* the slice thickness and do not count against this clearance — they are invisible to the transducer. - All 12 spiral wires (at canonical radii {4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26} mm) resolve clearly on the console before recording. Inner wires ( $r \leq 8$  mm) may sit close to the catheter ring-down; if a wire is consistently lost in ring-down at every azimuth, log it as `excluded_from_c_fit` in the metadata rather than chasing it with extra gain.
7. Allow  $\geq 60$  s for the milk to settle around the wire-phantom frame (the immersion will stir up some bubbles).
8. **Set the Phase B gain** per the wire-visibility rule above (deepest wire  $\geq 20$  dB above noise; shallowest wire  $\leq 95$  % saturated). For 30  $\mu$ m tungsten on the PV .035, expect to land at **gain 25–35**, not the Phase A reference of 50 — tungsten gives a strong specular reflection that saturates by gain  $\sim 40$  on the inner wires. Log the chosen gain value in



metadata as `phaseB_gain_slider`. Capture **30 frames** at that gain with the wire phantom in place.

9. (Fallback only — if the wire phantom is unavailable) Suspend a single 0.05 mm tungsten wire vertically through the milk at a measured radial distance (10–15 mm from the catheter centerline). Calliper-measure the distance both before and after the 30-frame capture and log as `r_callipered_mm_pre` / `r_callipered_mm_post`.

10. Remove the wire phantom; rinse with warm water + mild detergent to remove milk residue (milk fat will gum the wires within hours if left wet). Set aside on a clean towel — it will be re-used in Phase B2.

11. Log `T_milk` at the end of Phase B1.

## Fluid swap — undiluted to 1:1 diluted

12. Carefully drain the undiluted milk from the container into a discard vessel (do not lift the catheter; pour around it, or use a kitchen-grade pipette / turkey baster). Wipe the inside of the container with a clean paper towel — residual milk film is OK, but no visible droplets / pools.

13. Rinse the container with ~100 mL of the room-temperature degassed water from step 0; drain the rinse into the discard vessel. (This flushes the milk-fat film off the walls so it doesn't bias the diluted-fluid  $\alpha$ .)

14. Measure **250 mL evaporated milk** into the measuring jug. Add **250 mL room-temperature degassed water** (from step 0) on top. Gently swirl with a clean glass rod or wooden skewer to mix — do not shake (shaking entrains air; the dilution-induced bubbles take  $\geq 30$  min to settle).

15. Pour the 500 mL diluted-milk mixture into the container around the still-mounted catheter. Verify the fill height is  $\geq 80$  mm above the floor and the element ring has  $\geq 30$  mm clearance below the fluid surface; if short, add more diluted-milk mixture in the same 1:1 ratio.

16. Cover and let settle for  $\geq 30$  min (longer than for undiluted milk — the dilution stirs in fine bubbles that need extra time to rise out). Log `T_room` and `T_milk_diluted` at the end of the settle.

## Phase A2 — multi-gain diluted-milk capture

17. Repeat steps 1–4 (Phase A1) verbatim, in the same P1, P2, P3 positions, in the diluted milk. **Do not** re-mount the catheter between A1 and A2 — the analysis script cross-checks the TGC and  $\alpha$  extractions at the same axial positions across the two dilutions.

18. Log `T_milk_diluted` at the end of Phase A2.

## Phase B2 — wire-phantom c-refinement in diluted milk

19. Repeat steps 6–11 (Phase B1) verbatim, in the diluted milk. **Use the same operator-set gain as Phase B1** (logged as `phaseB_gain_slider`) — the wires have not moved and tungsten's saturation behaviour does not change between the two dilutions. Expected `c_diluted`  $\approx 1525$  m/s (vs  $\sim 1565$  m/s in B1); per-wire `r_displayed_k` for all 12 wires should shift by  $\approx 1.3\%$  between the two phases (the wires haven't moved; only the bulk  $c$  has changed). This is the TOF-extraction-pipeline cross-check.

20. After Phase B2, remove the wire phantom; rinse with warm water + mild detergent **before storing** (milk fat will gum the wires within hours if left wet).

21. Log `T_milk_diluted` at the end of Phase B2.

## Data to acquire

In what follows, `<sid_und>` is the undiluted substrate ID (`tmm-milk-INTERIM-<YYYYMMDD>-undiluted`) and `<sid_dil>` is the diluted ID (`tmm-milk-INTERIM-<YYYYMMDD>-dil1to1`).

### Phase A1 (undiluted):

- `t1_e4i_<sid_und>_phaseA1_p1_gain20.dcm`
- `t1_e4i_<sid_und>_phaseA1_p1_gain35.dcm`
- `t1_e4i_<sid_und>_phaseA1_p1_gain50.dcm`
- `t1_e4i_<sid_und>_phaseA1_p1_gain68.dcm`
- `t1_e4i_<sid_und>_phaseA1_p2_gain50.dcm`
- `t1_e4i_<sid_und>_phaseA1_p3_gain50.dcm`

#### Phase B1 (undiluted + wire phantom):

- `t1_e4i_<sid_und>_phaseB1_wirephantom_gainXX.dcm` (where `XX` is the operator-set gain from the Phase B gain rule, also logged as `phaseB_gain_slider` in metadata)

#### Phase A2 (diluted):

- `t1_e4i_<sid_dil>_phaseA2_p1_gain20.dcm`
- `t1_e4i_<sid_dil>_phaseA2_p1_gain35.dcm`
- `t1_e4i_<sid_dil>_phaseA2_p1_gain50.dcm`
- `t1_e4i_<sid_dil>_phaseA2_p1_gain68.dcm`
- `t1_e4i_<sid_dil>_phaseA2_p2_gain50.dcm`
- `t1_e4i_<sid_dil>_phaseA2_p3_gain50.dcm`

#### Phase B2 (diluted + wire phantom):

- `t1_e4i_<sid_dil>_phaseB2_wirephantom_gainXX.dcm` (same operator-set `XX` as Phase B1; logged as `phaseB_gain_slider`)

Each DICOM is 30 frames per (position, gain, dilution) combination. If the console concatenates captures, label the segments in the filename.

#### Metadata:

- `t1_e4i_<session_id>_metadata.json` — one per session, covering both dilutions. Fields: phase-A gain slider values (`phaseA_gain_sliders`: [20, 35, 50, 68]), **phase-B gain slider** (`phaseB_gain_slider`: `<operator-set value>`), imaging diameter, TGC schedule, acoustic-reference state, `T_room`, `T_milk_undiluted_start/end`, `T_milk_diluted_start/end`, evaporated milk brand / can lot / best-by date, **mL milk in each phase** (the as-poured amount, not the nominal 250 mL), **mL degassed water in the diluted phase**, degassed-water prep time + cooling time, container ID / dimensions, **wire phantom S/N + wire material + design radii** (the canonical 12-wire spiral, or fallback wire diameter and calliper distance), `excluded_from_c_fit` list recording any wires that were lost in catheter ring-down at every azimuth and therefore dropped from the c-extraction fit, photos of mount before and after the fluid swap.
- (Optional) `t1_e4i_<sid_*>_<segment>_polar.npy` — vendor polar-domain export, per segment.

## Analysis

Implementation should live in `instrument-calibration/p035_visions/extract_t1_e4i_milk.py` (one script per interim experiment, parallel to the canonical `tier1_evaluation.py`). The analysis layers the multi-gain and wire-phantom data from both dilutions into two output groups: **per-dilution outputs** (computed twice, once per `<sid_*>`) and **cross-dilution outputs** (computed once across the combined A1 + A2 / B1 + B2 data).

All per-dilution outputs are computed twice (once for `<sid_und>` from Phase A1+B1 and once for `<sid_dil>` from Phase A2+B2); cross-dilution outputs are computed once across the combined data.

## Per-dilution outputs (computed twice — once per `<sid_*>`)

| Output                                                                         | Source data                                                                                                          | Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mean intensity vs depth <code>I_dB(z; gain, position)</code>                   | Phase A1 or A2, all 6 captures                                                                                       | Convert each frame to polar B-mode, mask out the inner 0–2 mm (ring-down zone) and the outer 0.5 mm (boundary echoes), compute mean intensity per radial bin, convert to dB, average across the 30 frames of each capture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Interim</b><br><code>materials[].attenuation_db_per_cm</code><br>prior      | Phase A1/A2, P1 at gain 50 (avoid the gain-20 noise floor and gain-68 saturation; gain 35 used as a cross-check fit) | Fit $\text{slope\_dB\_per\_mm} = (I\_dB(z_2) - I\_dB(z_1)) / (z_2 - z_1)$ over $r \in [5, 25]$ mm at gain 50, and cross-check with the same fit at gain 35 — the two should agree to within $\pm 0.05$ dB/cm/MHz. $\alpha$ (dB/cm/MHz) = $\text{slope\_dB\_per\_mm} \times 10 / (2 \times f_0\_MHz)$ . Expected literature anchors: <b>0.6–1.0 dB/cm/MHz for <code>&lt;sid_und&gt;</code></b> (Antoniou 2021, undiluted evap milk $\approx 0.8$ dB/cm/MHz with brand/lot variation), <b>0.3–0.5 dB/cm/MHz for <code>&lt;sid_dil&gt;</code></b> (Farrer 2015 interpolated, 1:1 dilution $\approx 0.4$ dB/cm/MHz). Source field: <code>bench, milk-INTERIM-&lt;date&gt;-&lt;dilution&gt;</code> , no substitution measurement. Uncertainty $\pm 0.15$ dB/cm/MHz per dilution.                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Interim refined</b><br><code>probe.speed_of_sound_mm_per_us</code>          | Phase B1 or B2, wire phantom @ <code>phaseB_gain_slider</code> (operator-set; typically 25–35 for tungsten)          | For each of the (up to) 12 wires at the canonical Archimedean-spiral radii <code>r_design_k</code> $\in \{4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26\}$ mm, segment the wire echo (Hough or matched filter), read the polar-displayed radius <code>r_displayed_k</code> . Exclude any wire flagged in <code>excluded_from_c_fit</code> (typically the innermost 1–2 if they're lost in ring-down). Compute $c\_k = c\_displayed \times (r\_design\_k / r\_displayed\_k)$ , then take a weighted mean across the included wires (weight = inverse of localization uncertainty per wire, with extra down-weighting for the innermost 2 wires where ring-down can broaden the peak). <code>c_displayed</code> = $1.54 \text{ mm}/\mu\text{s}$ is the displayed-speed-of-sound assumption. Expected: <b><code>c_undiluted</code> <math>\approx 1560\text{--}1580 \text{ m/s}</math>, <code>c_diluted</code> <math>\approx 1515\text{--}1535 \text{ m/s}</math></b> . Uncertainty $\pm 2 \text{ m/s}$ per dilution (better than the original $\pm 3 \text{ m/s}$ with 5 wires because the 12-wire average gives $\sqrt{(12/5)} \approx 1.5\times$ more statistical power). |
| <b>Interim cross-check</b><br><code>processing.noise.sigma</code> (partial E5) | Phase A1/A2, P1 at gain 20                                                                                           | At gain 20 the deep-field ( $r \in [25, 28]$ mm) of plain milk is at or near the console's noise floor. Fit <code>scipy.stats.rayleigh.fit</code> to the envelope of the deep ROI; $\sigma$ is the noise $\sigma$ . Reported per dilution; the undiluted-milk $\sigma$ should be the cleaner electronic-noise-floor estimate (less signal contamination), the diluted $\sigma$ provides a higher-scatter cross-check.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Saturation diagnostic</b>                                                   | Phase A1/A2, P1 at gain 68                                                                                           | Count fraction of pixels in <code>r</code> $\in [3, 8]$ mm (near-field, highest intensity) that are within 1 palette unit of the maximum. If $> 1\%$ , the gain-68 capture saturated — flag the gain-68 row of the gain-linearity output as <code>saturated: true</code> and exclude that knot from the slope fit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Replicate stability                                                            | Phase A1/A2, gain 50 across P1/P2/P3                                                                                 | Compute <code>I_dB(z)</code> per axial position; check pairwise RMS deviation across positions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Cross-dilution outputs (computed once across A1+A2)

| Output                                                                            | Source data                                                                  | Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interim</b> <code>tgk_control_points</code> (final)                            | Phases A1 + A2, P1+P2+P3 at gain 50 across both dilutions (180 frames total) | The console TGC curve $g(z)$ is fixed across substrates. Solve $g(z) = -I_{dB}(z) + I_{dB}(0) + 2 \cdot \alpha \cdot z$ jointly across A1 ( $\alpha = \alpha_{undiluted}$ ) and A2 ( $\alpha = \alpha_{diluted}$ ); the per-dilution $\alpha$ values are taken from the per-dilution analysis above. Sample at depths {0, 1.5, 3.0, 4.5, 6.0, 8.0, 10.0, 15.0, 20.0, 25.0} mm (extended from canonical E4 to cover the 60 mm imaging diameter). The two dilutions must agree on $g(z)$ to within $\pm 2$ dB at every knot — otherwise the per-dilution $\alpha$ priors are off, or the pipeline has a depth-dependent bias. Source field: <code>bench, milk-INTERIM-&lt;date&gt;, cross-dilution joint fit, no substitution measurement</code> . |
| <b>Interim cross-check</b><br><code>gain_db(slider)</code> linearity (partial E7) | Phases A1 + A2, P1 at gains 20, 35, 50, 68 (8 captures total)                | Pick a stable mid-field radial band $r \in [10, 14]$ mm outside ring-down and clear of any saturation. Compute <code>mean_dB(gain_k) - mean_dB(gain=20)</code> for $k \in \{35, 50, 68\}$ for each dilution; the slider→dB transfer is a probe property and must be identical across dilutions. With 4 knots we can also fit a quadratic to detect bow (deviation from linear $\geq 2$ dB at any knot $\Rightarrow$ flag the slider→dB transfer as non-linear). Report <code>(slider, delta_dB)</code> table per dilution and the cross-dilution agreement; report mismatches $\geq 2$ dB as failures. Uncertainty $\pm 1$ dB per knot.                                                                                                          |
| <b>Pipeline self-consistency:</b><br>$\alpha$ -vs-milk-fraction linearity         | $\alpha$ priors from both dilutions                                          | Compute the ratio $\alpha_{undiluted} / \alpha_{diluted}$ . Expected: <b><math>2.0 \pm 0.2</math></b> (Antoniou 2021 reports linear $\alpha$ scaling with milk fraction over 10–100 % milk). A ratio $< 1.6$ or $> 2.4$ indicates either a non-linear $\alpha$ response (potential brand issue) or a pipeline bias in $\alpha$ extraction. Reported as a pass/fail flag plus the actual ratio.                                                                                                                                                                                                                                                                                                                                                   |
| <b>Pipeline self-consistency:</b> TGC recovery agreement                          | $g(z)$ from per-dilution Phase A1 and A2 independent fits                    | For diagnostic purposes, also fit $g(z)$ separately from A1 alone and A2 alone (each using its own $\alpha$ prior). The two independent $g(z)$ curves should agree to within $\pm 2$ dB across the 0–30 mm range. Larger disagreement is the canary for an $\alpha$ -prior or TGC-fit bias.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pipeline self-consistency:</b><br>wire-phantom $\Delta c$                      | $c$ from B1 vs $c$ from B2                                                   | Expected: <code>c_undiluted - c_diluted</code> $\approx 40 \pm 15$ m/s (undiluted milk is denser + faster than the 1:1 mix; the actual $\Delta c$ depends on the milk's total solids fraction, but the <i>sign</i> must be positive). A negative or null $\Delta c$ indicates either a TOF-extraction bug or that one of the two dilutions isn't what it's labelled as.                                                                                                                                                                                                                                                                                                                                                                          |

## Bench acceptance criteria

Per-dilution criteria (must pass independently for both `<sid_und>` and `<sid_dil>`):

- Per-axial-position mean palette in any 5 mm radial bin stable to within **±3 palette units** across the three gain-50 replicates (looser than canonical E4's ±2 because milk has slightly more inter-position variation than gel TMM).
- $\alpha$  extracted from gain-50 `I_dB(z)` slope falls within the **per-dilution literature window**: 0.6–1.0 dB/cm/MHz for `<sid_und>`, 0.3–0.5 dB/cm/MHz for `<sid_dil>`. Values outside the appropriate band indicate either a temperature drift, a fluid-mix error (e.g. wrong volume ratio for the diluted phase), or a TGC-not-flat capture error.
- `T_milk` start vs end stable to within **±1 °C** during each phase (A1, B1, A2, B2 logged independently).
- Gain-linearity `delta_dB(gain=68) - delta_dB(gain=50)` is in the range `[8, 22]` dB (the gain-50→68 step is canonically ~15 dB on this console; outside this band means either saturation or a console setting changed mid-capture). Must hold for **both** dilutions.
- Gain-linearity `delta_dB(gain=35) - delta_dB(gain=20)` is in the range `[8, 22]` dB (the gain-20→35 step should be similar to the 50→68 step — both are 15 slider units; large asymmetry between the two indicates bow in the slider→dB transfer that must be captured by the quadratic fit rather than ignored).
- Gain-68 saturation diagnostic reports < 5 % saturation; if higher, drop the gain-68 row and re-run the linearity fit on just (20, 35, 50).
- Gain-35 saturation diagnostic (same method, applied to gain 35) reports 0 % saturation — gain 35 should never saturate even in the near-field; saturation here indicates a TGC slider drift or a brighter-than-expected substrate.
- Phases B1/B2: per-wire `r_displayed_k` consistent across the 30 frames to within ±0.05 mm; per-wire `c_k` values agree across the **included** wires (those not in `excluded_from_c_fit`) to within **±10 m/s** within each dilution; at least **8 of 12 wires** must contribute to the c fit (if more than 4 wires are lost in ring-down or saturation, lower the Phase B gain and re-acquire).

Cross-dilution criteria (one pass/fail flag each across A1+A2 / B1+B2):

- $\alpha$  **ratio** `alpha_undiluted / alpha_diluted` falls in **2.0 ± 0.4** (loose Antoniou 2021 linearity bound; ±0.4 accounts for ±0.15 uncertainty in each  $\alpha$ ). Failure here is the loudest red flag — it means either the dilution was wrong (mismeasured volumes) or the  $\alpha$ -extraction pipeline has a bias.
- **TGC self-consistency**: per-dilution `g(z)` independent fits agree to within ±2 dB at every depth knot.
- **Gain-linearity self-consistency**: `delta_dB` per slider step matches between dilutions to within ±1.5 dB (probe property — must not depend on substrate).
- **$\Delta c$  sign + magnitude**: `c_undiluted - c_diluted` is positive and in the range **15–80 m/s**.

If any per-dilution criterion fails, re-run *that dilution* with a fresh can / fresh dilution mix. If a cross-dilution criterion fails while per-dilution criteria pass, investigate the analysis pipeline before re-running — the data may still be usable.

## Provenance for the substrate parameter sheet

T1-E4\* generates **two** substrate-parameter-sheet rows per session (one per dilution). Each row must include:

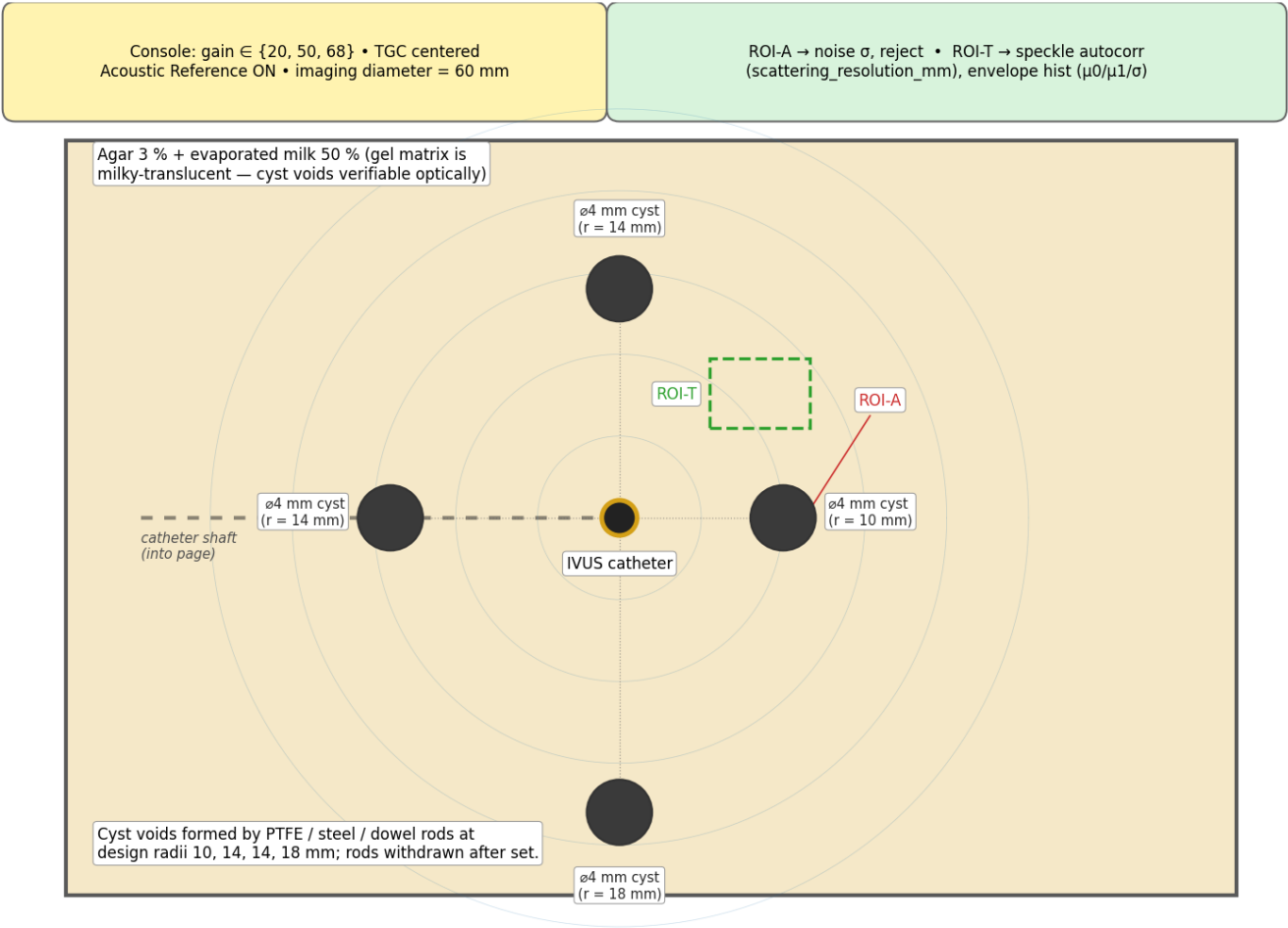
- Source: bench, interim milk batch `<YYYYMMDD>-<dilution>`, no substitution measurement
- Uncertainty: ±0.15 dB/cm/MHz ( $\alpha$ ) / ±3 m/s (c, wire phantom) / ±1 dB per knot (gain\_db cross-check) / ±10 % (noise  $\sigma$  cross-check)
- Notes: TIER 1 INTERIM PRIOR — re-derive with canonical agar-graphite phantom (A.2) when available; gain\_db and noise  $\sigma$  are CROSS-CHECK values, the authoritative versions still come from T1-E7 and T1-E5\* respectively;  $\alpha$  cross-validated across `<sid_und>` and `<sid_dil>` with ratio = `<reported>:1`

The **TGC schedule** (from the cross-dilution joint fit) is recorded as a *single* row, since the console TGC is substrate-invariant:

- Source: bench, interim milk batch <YYYYMMDD>, cross-dilution joint fit, no substitution measurement
  - Uncertainty:  $\pm 2$  dB per knot (cross-dilution joint-fit residual)
  - Notes: TIER 1 INTERIM PRIOR – recovered jointly from <sid\_und> and <sid\_dil>; per-dilution independent fits agree to within <reported> dB; re-derive with canonical agar-graphite phantom (A.2) when available
-

# T1-E5\* — Agar + evaporated-milk cyst phantom (interim stand-in for E5)

## T1-E5\* — Agar + evaporated-milk cyst phantom (interim stand-in for E5; cross-section view)



**Produces:** interim processing.noise.type, processing.noise.sigma, processing.scattering\_resolution\_mm, processing.scatter\_integral\_scale, processing.reject\_db, and priors on per-tissue scattering parameters (mu0, mu1, sigma).

**Substrate ID pattern:** tmm-milk-agar-INTERIM-<YYYYMMDD>.

**Goal.** Embed PTFE-rod-formed cyst voids in a milk-agar gel to reproduce the canonical E5 geometry (catheter at center, four Ø4 mm cysts at design radii 10, 14, 14, 18 mm). The gel matrix is milky-translucent (not opaque black like graphite-loaded agar), which has the side benefit of making the canonical A.3 optical cyst-diameter acceptance check actually achievable.

## Equipment

|               |                                              |              |
|---------------|----------------------------------------------|--------------|
| Equipment     | Part Number                                  | Manufacturer |
| Agar          | Agar 3 % + evaporated milk 50 %              | Agar         |
| PTFE rods     | PTFE rods at design radii 10, 14, 14, 18 mm  | PTFE         |
| Steel rods    | Steel rods at design radii 10, 14, 14, 18 mm | Steel        |
| Dowel rods    | Dowel rods at design radii 10, 14, 14, 18 mm | Dowel        |
| IVUS catheter | IVUS catheter                                | IVUS         |
| ROI-T         | ROI-T                                        | ROI-T        |
| ROI-A         | ROI-A                                        | ROI-A        |

## Recipe — milk-agar-glycerin gel (interim)

This is a drop-in modification of the canonical A.2 recipe: the milk replaces the graphite-scatterer role, glycerin replaces n-propanol as the sound-speed adjuster (no shelf-life decay, food-grade, and already cited as the IEC 60601-2-37 / Souza 2016 replacement in Tier 2 protocol Appendix B.2), and Tween-20 is omitted because there's no graphite to wet.

| Component                                                                                                      | Mass                                                       | Function                                                | Tolerance                                                                                           |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Distilled water                                                                                                | 76 g                                                       | Gel solvent                                             | ± 1 g                                                                                               |
| Evaporated milk<br>(Carnation-type, ~25 % total solids)                                                        | 100 g (≈ 95 mL of one can)                                 | Scatterer + $\alpha$ contributor<br>(replaces graphite) | ± 1 g; do not exceed 50 %<br>by mass of the liquid                                                  |
| <b>Glycerin (≥ 99.5 %, USP / food / "vegetable glycerine" grade — single-ingredient)</b>                       | <b>24 g (12 % w/w of total)</b>                            | Sound-speed adjuster (target $c \approx 1540$ m/s)      | ± 0.5 g; do <b>not</b> substitute glycerin blends (e.g. "glycerin + rosewater")                     |
| Agar (high-strength, ≥ 600 g cm <sup>2</sup> gel strength, single-ingredient food-grade or microbiology-grade) | 6 g (≈ 2.7 % w/w of the liquid phase; ≈ 3 % of water+milk) | Gel matrix                                              | ± 0.05 g; bump to 6.7 g if the demolded gel feels too soft to hold the cyst rods (rare)             |
| Potassium sorbate (food / USP grade)                                                                           | 0.4 g (0.2 % w/w)                                          | Preservative                                            | ± 0.05 g                                                                                            |
| <del>n-Propanol</del>                                                                                          | omit                                                       | (would also set $c$ to 1540 m/s)                        | Replaced by glycerin in this recipe; either is acceptable but glycerin is much better on shelf life |
| <del>Tween-20</del>                                                                                            | omit                                                       | (would wet graphite)                                    | Milk scatterers don't need surfactant                                                               |
| <del>Graphite</del>                                                                                            | omit                                                       | (would be the primary scatterer)                        | Milk fat globules + casein micelles replace                                                         |

Theoretical properties (literature-prior + Urick mixing-rule estimate for the milk+water+glycerin liquid phase, will refine per batch):

- **$c \approx 1535\text{--}1545$  m/s** (Urick mixing-rule estimate for the 76 g water / 100 g milk / 24 g glycerin liquid phase, plus the ~+5 m/s contribution of the agar matrix; brought up from the no-glycerin recipe's ~1510 m/s).
- **$\alpha \approx 0.3\text{--}0.5$  dB/cm/MHz at 10 MHz** (lower than the 0.5 dB/cm/MHz canonical target because the milk solids are diluted by the agar matrix; glycerin adds a small ~+0.02 dB/cm/MHz contribution which is in the noise; substitute-measure per A.6 if you can run one this week).
- **$Z \approx 1.58$  MRayl** (glycerin nudges density up by ~1.5 % over the no-glycerin recipe).
- **Gel firmness:** equivalent to a 3 % graphite-loaded agar; pourable at 50 °C, holds PTFE rods, demolds cleanly. Glycerin slightly plasticizes the gel — that's a feature, not a bug, for our use (less brittle than pure agar; the canonical A.2 phantom is known to fracture on rough handling).
- **Shelf life:** glycerin is hygroscopic and non-volatile, so the  $c$ -drift mechanism that limits canonical-A.2 to ~14 days (propanol evaporation through the gel surface) does not apply here. Milk-protein bacterial growth is now the limiting factor, which the potassium sorbate addresses; expect **7–10 days at 4 °C** with the gel wrapped in cling film inside a



sealed bag. The canonical 5–7 days figure (pre-glycerin draft) can be relaxed by ~2 days because glycerin itself is mildly bacteriostatic above 10 % w/w.

## Preparation procedure

The key delta from canonical A.2 is **add milk after the agar is dissolved and the slurry is below 70 °C** — milk proteins denature and Maillard-brown above 80 °C, which changes the speckle properties and discolors the gel.

1. **Prepare the mold (cyst-rod registration).** Lightly oil the mold interior with a release agent (silicone spray).

(a) *Preferred — printed mold + lid:* Use `hardware/phantom_mold_cyst_interim.stl` (interim 8 mm cysts at shifted radii) or `phantom_mold_cyst.stl` (canonical 4 mm cysts) paired with the matching `..._lid_interim.stl` / `..._lid.stl`. Wipe the mold interior with a thin coat of food- grade silicone spray or cooking-oil spray; let any beads of excess oil drain out before proceeding.

(i) **Seat the catheter rod.** Take your channel-forming rod (Ø 3 mm wooden skewer, PTFE rod, etc. — see equipment table) and coat it with the same release agent: spin it through an oiled palm or paper towel and wipe off the excess so a uniform thin film remains. Drop the rod into the mold's centre bore (Ø 3.5 mm, passes through the gel cavity into a 2 mm-deep blind divot in the floor — the rod's bottom seats in the divot and the floor stays solid below it, so no gel can leak out during pouring). The rod should drop in with light force and stand vertical under its own weight; if it binds, ream the bore with a 3.5 mm drill bit.

(ii) **Seat the cyst rods.** Lightly oil each of the four cyst rods. Press them firmly into the floor dimples at the design radii — **{12, 16, 16, 20} mm for the 8 mm rigid plastic straws**, or **{10, 14, 14, 18} mm for the canonical 4 mm PTFE / steel rods**. They should stand vertical without external support; if they wobble, the dimples may need a quick clean-out with a drill bit of the matching diameter.

(iii) **Seat the lid (mechanical key alignment).** Place the matching registration lid (`..._lid_interim.stl` or `..._lid.stl`) over the rod tops so the catheter rod and all 4 cyst rods pass cleanly through the lid's pre-drilled through-holes. The lid has a rectangular **orientation key tab** on its underside at 0° (5 mm wide × 3 mm tall, extending from the underside of the top plate at the wall position). Rotate the lid until this tab drops into the matching slot in the mold rim at 0° — when correctly oriented, the lid seats flush on the rim with no perceptible play; **if the lid sits visibly proud of the rim by ~3 mm, the tab is over the wall, not the slot — lift the lid and rotate until it drops**. With the lid fully seated, both ends of every rod are mechanically registered (lid hole at the top, floor divot/dimple at the bottom) and the rods are forced vertical — no external clamps are needed. Stand the assembly on a level surface.

(b) *Fallback — plain food-storage container + register card:* If the printed mold + lid is not available, cut a stiff "register card" from any flat, food-safe-adjacent material — thin acrylic, the side panel of a metal milk can (folded 2× for stiffness), 4-layer-folded aluminum foil, or stiff cardboard — sized to lie flat on a 9 × 9 × 8 cm food-storage container's top rim. Punch 5 holes through the register card: - 1 center hole, Ø 2 mm, for the PTFE catheter channel rod; - 4 cyst-rod holes, sized to the cyst-rod diameter, at design radii **{10, 14, 14, 18} mm if using 4 mm cysts** or **{12, 16, 16, 20} mm if using 8 mm cysts** (the shifted radii keep the innermost cyst's inner edge ≥ 8 mm from the catheter, clear of ring-down). The angular layout is always {0°, 90°, 270°, 180°}.

Tape the register card to the container's top rim. Push the 5 rods (1 catheter + 4 cyst) through the card so they hang vertically inside the container, with their lower tips touching the container floor. Anchor each rod's lower tip with a pea-sized ball of **food-grade plumber's putty or Crayola modeling clay** on the container floor — both are removable, grippy on plastic, and release cleanly when the rod is pulled. Verify each rod is vertical with a small bubble level on top of the register card. Stand the assembly on a level surface. 2. **Hydrate the agar in pure water (NOT milk).** Stir 6 g agar into **76 g** of cold distilled water in the beaker; rest 5 min to fully wet. Agar dissolves much better in clean water than in milk or in a glycerin solution. 3. **Heat to dissolve agar.** Bring the agar-water slurry to a gentle boil on the hot plate with continuous magnetic stirring. Hold at  $90 \pm 2$  °C for **5 minutes** until you see no agar granules and the liquid is clear amber. 4. **Cool to 70 °C with continuous stirring.** Take off the hot plate if needed; the slurry should cool from 90 to 70 °C in ~3 min. *Do not let it fall*

below 60 °C before the next step. 5. **Add the glycerin.** Pour the 24 g of glycerin into the agar solution. Glycerin is fully miscible with water and won't shock the agar. Stir for **2 minutes** to homogenize. The temperature will drop a couple of °C; keep above 60 °C. 6. **Add the evaporated milk.** Pour the 100 g of room-temperature evaporated milk into the agar-glycerin solution. The temperature should drop to ~50–55 °C immediately. Stir gently for 2 minutes to homogenize. **Do not reheat above 70 °C after the milk is added** — that's what causes Maillard browning and protein denaturation. (Adding the glycerin *before* the milk also slightly reduces the temperature shock to the milk proteins, because the glycerin-water mixture has a higher heat capacity per gram than plain water.) 7. **Add potassium sorbate.** Stir in the 0.4 g of K-sorbate. 8. **Degas.** Pour into the vacuum chamber and pull  $\geq 70$  kPa for 5 min, or sonicate in the ultrasonic bath for 10 min, until visible micro-bubbles are gone. Milk-agar-glycerin foams more easily than pure agar-graphite (glycerin reduces surface tension slightly), so this step is more important than usual — allow the full 5 min even if the foam looks gone at 2 min. 9. **Pour the mold.** While still at 45–50 °C, pour into the prepared mold in one continuous pour. Avoid splashing onto the mold walls above the fill line. Tap the mold gently to release wall-bubbles. 10. **Set at 4 °C for  $\geq 4$  h.** Cover with cling film. The gel firms up similarly to pure agar; glycerin slows the set time slightly (allow 4–5 h instead of 3–4 h). 11. **Demold the cyst-forming rods.** Twist + vertical pull on each of the four cyst rods (PTFE: easy; steel: light twist+pull; dowel: gentle twist+pull, may shed a small amount of fiber); vertical pull on the catheter channel rod. Glycerin makes the rods marginally easier to release because the gel surface stays slightly tacky. 12. **Storage.** Wrap in cling film + sealed bag at 4 °C. **Use within 7–10 days** — glycerin + potassium sorbate together hold the milk-protein-bacterial-growth limit out to ~10 days, vs the canonical graphite-agar phantom's 14 days per A.2.

### Mold acceptance criteria (analogous to A.3)

- All 4 cyst voids cleanly demolded; walls intact, no tears or collapsed inclusions.
- Each cyst diameter at the imaging plane within  $\pm 0.2$  mm of nominal (canonical 4.0 mm for PTFE/steel rods; **8.0 mm for the rigid reusable plastic-straw option** — adjust the analysis ROI sizes accordingly). **Measured optically through the gel** with a USB microscope + graticule card behind the phantom: the milk-agar gel is translucent enough at 50 % milk content that this measurement is actually feasible.
- Cyst-to-catheter radial distance at the imaging plane within  $\pm 0.3$  mm of design — {10, 14, 14, 18} mm for 4 mm cysts or **{12, 16, 16, 20} mm for 8 mm cysts** — measured with the same optical setup.
- Visible homogeneity: no settling layer of milk solids at the bottom (a thin  $< 0.5$  mm cream layer at the top is acceptable and won't affect the imaging plane).
- Mass after demold within  $\pm 2$  % of the design mass ( $\approx 206.4$  g = 76 + 100 + 24 + 6 + 0.4) — confirms full liquid was retained.
- (If substitution-attenuation measured per A.6)  $\alpha$  within 0.3–0.6 dB/cm/MHz; record the measured value as the substrate-sheet  $\alpha$  and note *Source: bench, substitution per A.6, interim milk-agar-glycerin batch <date>*.

### Setup procedure (imaging session)

1. Mount the phantom in the imaging plane with the catheter channel accessible. Verify the catheter channel is clear and dry; insert the catheter to the marked imaging plane (typically 30 — 40 mm into the channel).
2. Allow  $\geq 60$  s of equilibration after catheter movement.
3. Log T\_phantom (insert a thermometer through a small hole in the gel surface, away from the catheter and cysts).

### Acquisition procedure

Capture **30 frames at static position** at **each of three gain settings**: gain 20 (low; clean noise floor), gain 50 (reference operating point), gain 68 (high; pushes saturation). Repeat the acquisition at the **same** catheter position for all three gains.

Total: 3 gains × 30 frames = 90 frames at one position.

## Data to acquire

- `t1_e5i_<substrate_id>_gainXX.dcm` for  $XX \in \{20, 50, 68\}$  — one DICOM per gain, 30 frames each.
- `t1_e5i_<substrate_id>_metadata.json` — gain settings, imaging diameter, TGC schedule, AR state, T\_phantom, milk batch / best-by-date, agar batch, cyst-rod material(s), measured cyst diameters and radial distances (from the mold acceptance check), photos of mount.
- `t1_e5i_<substrate_id>_rois.json` — ROI definitions:
  - `roi_A`: anechoic interior of one well-formed cyst ( $\geq 3 \times 3 \text{ mm}^2$  of cyst lumen, no boundary). The figure suggests using the  $r=10 \text{ mm}$  cyst as ROI-A but any of the four works as long as the ROI is fully inside the cyst.
  - `roi_T`: uniform-gel region ( $\geq 5 \times 5 \text{ mm}^2$ , away from any cyst, the catheter ring-down zone, or the mold wall).
- (Optional) `t1_e5i_<substrate_id>_polar.npy` — vendor polar export.

## Analysis

Implementation lives in `instrument-calibration/p035_visions/extract_t1_e5i_milk_agar.py`.

| Output                                                                  | Method                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interim <code>noise.sigma</code> (linear units)                         | At gain = 20, take an ROI in the deep gel field ( $r \in [25, 28] \text{ mm}$ ) where attenuation has driven speckle below the noise floor. Fit <code>scipy.stats.rayleigh.fit</code> to the envelope; $\sigma$ is the noise $\sigma$ . For envelope-only DICOM, use the same Rayleigh- $\sigma$ method.                                                                                                                                                      |
| Interim <code>noise.type</code>                                         | KS-test of the deep-ROI envelope histogram against Rayleigh / Gaussian / Nakagami. Pick the lowest-D distribution.                                                                                                                                                                                                                                                                                                                                            |
| Interim <code>reject_db</code>                                          | <code>dB_below_T_at_roi_A_50_pct</code> — the dB level at which 50 % of <code>roi_A</code> pixels fall below the display threshold.                                                                                                                                                                                                                                                                                                                           |
| Interim <code>scattering_resolution_mm</code>                           | 2D autocorrelation of <code>roi_T</code> ; report the 1/e radius along the radial and lateral axes (use mean for the scalar field).                                                                                                                                                                                                                                                                                                                           |
| Interim <code>scatter_integral_scale</code>                             | Run simulator with measured material defaults; tune until simulated <code>roi_T</code> mean matches measured <code>roi_T</code> mean within 1 dB.                                                                                                                                                                                                                                                                                                             |
| Interim <code>mu0</code> / <code>mu1</code> / <code>sigma</code> priors | Fit a homodyned-K distribution to the <code>roi_T</code> envelope histogram. Map shape parameter $k \rightarrow \mu_0/\mu_1$ (literature mapping); RMS amplitude $\rightarrow \sigma$ .                                                                                                                                                                                                                                                                       |
| Diagnostic — speckle autocorrelation length (sim vs bench)              | Same <code>roi_T</code> autocorrelation as above, but <b>also</b> render a uniform-tissue sim frame at the reference operating point with the current <code>volcano_s5i.yaml</code> , compute its autocorrelation length, and report ( <code>bench</code> , <code>sim</code> , <code>ratio</code> ). The ratio is the headline diagnostic for the "sim looks too sharp" observation. <b>This is the most informative output from T1-E5* — do not skip it.</b> |

## Bench acceptance criteria

- `roi_A` mean intensity drops by  $\geq 25$  dB vs `roi_T` (slightly looser than the canonical E5's 30 dB threshold because the milk-agar speckle floor is closer to the cyst interior than a graphite-loaded gel's; below 25 dB indicates a poorly-formed cyst or trapped bubbles).
- KS p-value for chosen noise distribution  $\geq 0.05$ .
- Speckle autocorrelation length consistent across gain settings ( $CV \leq 15\%$ ).
- Bench-vs-sim autocorrelation ratio reported (no pass/fail — diagnostic only at the interim stage).

## Provenance for the substrate parameter sheet

Every row populated from T1-E5\* must include:

- Source: bench, interim milk-agar batch <YYYYMMDD>, no substitution measurement (or ... substitution per A.6 if you ran it)
  - Uncertainty:  $\pm 10\%$  for noise  $\sigma / \mu / \text{sigma}$ ;  $\pm 15\%$  for `scattering_resolution_mm`
  - Notes: TIER 1 INTERIM PRIOR — re-derive with canonical agar-graphite cyst phantom (A.3) when available; speckle-autocorrelation diagnostic figure attached separately
-

## Reproducibility

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Schematic figures for this SOP are generated by the same tool as the canonical protocol figures:

```
PYTHONPATH=/path/to/matplotlib MPLBACKEND=Agg \  
python3 instrument-calibration/tools/gen_calibration_figures.py
```

The two SOP figures ([fig\\_t1\\_e4i\\_milk\\_tank.png](#) and [fig\\_t1\\_e5i\\_milk\\_agar\\_cyst.png](#)) live in [instrument-calibration/docs/calibration\\_figures/](#) alongside the canonical E1–E9 figures.

## Closing — when to retire this SOP

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Once the canonical agar-graphite phantoms (A.2 and A.3) are available — i.e. once 3  $\mu\text{m}$  flake graphite (bulk density  $\geq 0.3$  g/mL) arrives (glycerin is already on-hand from this SOP and can be carried over to the canonical A.2 as the n-propanol replacement) — the canonical E4 and E5 should be **re-run from scratch** and the substrate parameter sheet rows updated:

- Drop the **INTERIM** token from the `substrate_id`.
- Replace the **Source** field with the canonical phantom batch ID.
- Tighten the **Uncertainty** columns to the canonical-protocol values ( $\pm 0.10$  dB/cm/MHz for  $\alpha$ ,  $\pm 2$  m/s for  $c$ ,  $\pm 5$  % for the speckle statistics).
- Archive the interim DICOM / metadata files in `<bench_dataset>/interim_milk_phantom_<YYYYMMDD>/` rather than deleting them — they remain useful as cross-batch sanity-check references and as the source of the speckle-autocorrelation diagnostic figure.

The T1-E7 step-attenuation phantom has **no equivalent in this SOP**; it must wait for canonical graphite.

## References

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Foundational TMM literature (cited above for  $c$ ,  $\alpha$  targets and milk-as-TMM precedent):

**Madsen E.L., Zagzebski J.A., Banjavic R.A., Jutila R.E.** (1978). *Tissue mimicking materials for ultrasound phantoms*. Med. Phys. 5(5):391–394. doi:10.1118/1.594483 — Original water-based agar + graphite + n-propanol TMM. Sets the canonical  $\alpha = 0.5\text{--}1.0$  dB/cm/MHz target. This is the **agar-graphite** paper that backs canonical T1-E4/E5/E7; it is **not** a milk paper.

**Goss S.A., Johnston R.L., Dunn F.** (1978). *Comprehensive compilation of empirical ultrasonic properties of mammalian tissues*. J. Acoust. Soc. Am. 64(2):423–457. doi:10.1121/1.382016 · PubMed 361793 — Where the 1540 m/s and  $\sim 0.5$  dB/cm/MHz tissue targets come from.

**Goss S.A., Johnston R.L., Dunn F.** (1980). *Compilation of empirical ultrasonic properties of mammalian tissues. II*. J. Acoust. Soc. Am. 68(1):93–108. Free PDF · PubMed 11683186 — Follow-up compilation with 45 more tissue studies.

**Madsen E.L., Frank G.R., Dong F.** (1998). *Liquid or solid ultrasonically tissue-mimicking materials with very low scatter*. Ultrasound Med. Biol. 24(4):535–542. PubMed 9651963 — The WARF / Madsen evaporated-milk-based gel material, patented at U. Wisconsin and distributed in thousands of commercial CIRS / ATS phantoms. Establishes evaporated milk + congealing solution as a long-shelf-life soft-tissue TMM.

**Farrer A.I., Odéen H., de Bever J., Coats B., Parker D.L., Payne A., Christensen D.A.** (2015). *Characterization and evaluation of tissue-mimicking gelatin phantoms for use with MRgFUS*. J. Ther. Ultrasound 3:9. doi:10.1186/s40349-015-0030-y · PubMed 26146557 — **Gelatin + 50/50 water/evaporated-milk** phantom. Carnation evaporated milk (6.3 % fat). Reports  $c$ ,  $\alpha$ , density, Young's modulus, MR T1/T2\* across three bloom values. At 50/50 milk/water they report  $c \approx 1540$  m/s and  $\alpha \approx 0.4\text{--}0.5$  dB/cm/MHz, matching the soft-tissue targets.

**Antoniou A., Damianou C., et al.** (2021). *Ultrasonic Attenuation of an Agar, Silicon Dioxide, and Evaporated Milk Gel Phantom*. J. Med. Ultrasound 29(4):239–249. doi:10.4103/JMU.JMU\_145\_20 · PMC8772477 · PubMed 35127403 — **The closest published match to the T1-E5\* recipe in this SOP.** 9 agar-based phantoms varying agar (2–6 %), SiO<sub>2</sub> (2–6 %) and evaporated milk (10–30 %). Reports  $\alpha$  range 0.30–1.49 dB/cm/MHz scaling  $\sim$ linearly with milk fraction; whole homogenized milk  $\alpha \approx 0.35$  dB/cm/MHz; evaporated-milk  $\alpha$ -slope contribution  $\approx 0.8$  dB/cm/MHz; recommended soft-tissue phantom  $\alpha$  range 0.3–0.7 dB/cm/MHz.